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## Title

An Open Non-Invasive EEG Dataset for Spontaneous Auditory Attention Switch Decoding

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## Abstract

Auditory brain–computer interfaces (BCIs) based on non-invasive electrodes offer significant potential for advancing the understanding of selective auditory attention mechanisms and enabling natural human–computer interaction. Despite growing interest in auditory attention research, publicly available datasets focused on spontaneous auditory attention switching remain limited, particularly those with high-quality electroencephalography (EEG) recordings in realistic listening environments. To address this gap, we present the Auditory Attention Switching Dataset (AASD), a non-invasive EEG dataset designed to investigate spontaneous selective auditory attention switching during naturalistic auditory processing. The dataset captures both sustained attention and spontaneous attention switching events through EEG signals. A baseline decoding model is introduced to verify data integrity and demonstrate its potential for practical applications. This open-access resource lays the foundation for developing algorithms for spontaneous auditory attention switching and advancing research in natural-scenario auditory BCIs.

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## Background & Summary

Auditory BCIs enable the decoding of neural responses to external auditory stimuli, allowing for the identification and interpretation of selective auditory attention states. This capability has significant potential for naturalistic human–computer interaction, cognitive neuroscience, and auditory perception research[1-3]. Early research established the feasibility of neural-based attention tracking. For instance, the seminal work by O'Sullivan[4] demonstrated that attentional selection within a 'cocktail party' scenario could be robustly decoded from single-trial EEG, laying a critical foundation for real-world applications in dynamic acoustic environments. Complementing this, Mesgarani and Chang[5] revealed that selective cortical representations of attended speakers are reliably maintained during multi-talker perception, further validating the use of auditory attention as a viable control signal for next-generation Brain-Computer Interfaces (BCIs). To translate these cortical representations into practical assistive technologies, electroencephalography (EEG) has emerged as the primary recording modality. Compared with other neural signal acquisition techniques, EEG offers unique advantages including non-invasiveness, broad applicability, and excellent temporal resolution[6,7]. These properties make EEG particularly well suited for tracking dynamic attention in realistic listening environments.

Most existing auditory attention decoding studies have relied on passive listening or experimenter-

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controlled attention-switching paradigms[8-12]. While these approaches have played a crucial role in advancing decoding algorithms and feature extraction methods, they differ substantially from real-world listening conditions. In real scenarios, auditory attention is influenced by semantic cues, personal interests, or environmental distractions, leading to spontaneous and nonlinear switching between multiple sound sources[13-15]. Understanding the neural mechanisms underlying such spontaneous attention switching is essential for developing more robust and generalizable auditory attention decoding algorithms. However, publicly available datasets in this domain remain scarce, particularly those containing high-quality, long-duration EEG recordings.

To address this gap, we introduce and publicly release the AASD dataset[16], a non-invasive EEG dataset focusing on spontaneous selective auditory attention switching in real scenarios. In the experimental paradigm, participants simultaneously listened to two narrative audio streams, namely male and female voices, spatially presented via HRTF to the left and right ears. Throughout the entire process, they are allowed to freely choose the source of the sound that they wanted to focus on. When shifting their attention between streams, participants indicate their attention direction in real time through button presses, enabling the recording of multiple spontaneous attention switches within a single trial. This design allows researchers to capture attention as it evolves naturally, rather than through externally imposed task constraints. The dataset comprises recordings from 18 healthy participants, with a total of 1260 minutes of EEG data. The recordings are originally sampled at 500 Hz during acquisition and resampled to 128 Hz after preprocessing. Audio stimuli, behavioral markers, and EEG signals are precisely synchronized and temporally aligned to ensure accurate labeling of attention states. This results in continuous, high-quality neural recordings that cover both attention maintenance and spontaneous attention switching processes. In addition to the dataset, we provide a baseline auditory attention switching decoding model to support algorithm benchmarking, facilitate reproducibility and encourage methodological comparisons.

The primary goal of this work is to establish a high-quality, openly accessible research resource that supports the development of auditory attention switching studies under naturalistic conditions. By systematically recording and annotating neural signals associated with spontaneous selective auditory attention switching, the AASD dataset provides a valuable foundation for exploring the dynamic mechanisms of auditory attention. Furthermore, the high temporal resolution of this dataset makes it highly suitable for training, validating and comparing various neural decoding algorithms, and it also lays an important foundation for achieving more robust and generalized auditory BCIs.

## Methods

**Participants.** The participants in this study were 18 healthy volunteers, aged between 18 and 27 years. All participants were native Mandarin speakers with normal hearing and none reported a history of neurological or psychiatric disorders. All experimental sessions were conducted in a quiet laboratory environment with stable lighting conditions to minimize external interference and ensure consistency across participants. The study protocol and all experimental procedures were reviewed and approved by the Institution's Ethical Review Board of Southern University of Science and Technology (No. 2022DZX003). All methods were carried out in accordance with relevant guidelines and regulations. Prior to the experiment, the researchers provided a detailed explanation of the study procedures, task requirements, and data-sharing policy. Written informed consent was obtained from each participant, confirming their full understanding of the experimental procedures and voluntary participation. All participants also explicitly agreed to the public sharing of their anonymized data. To ensure privacy protection, all personally identifiable information was removed from the dataset, and verified that no

indirect identifiers remained that could enable re-identification. Before the formal experiment, all participants completed a standardized training session to ensure they fully understood the experimental tasks and attention-switching requirements.

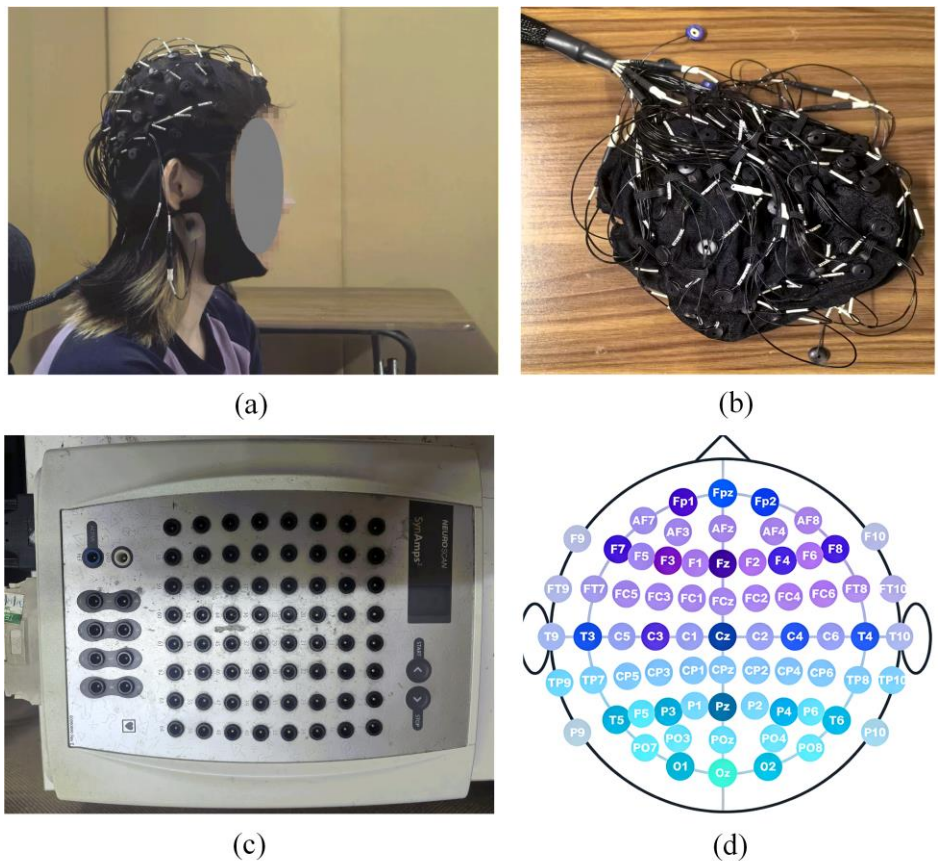


Fig. 1 EEG Data Acquisition System Architecture. (a) The EEG experimental setup, (b) the EEG cap with a 64-channel electrode configuration, (c) the 64-channel high-resolution EEG signal amplifier, and (d) the 64-channel 10-20 electrode layout diagram. The individual depicted in this figure has provided explicit written informed consent for the open publication of her image.

**EEG Data Acquisition.** EEG recordings were conducted in a double-walled, acoustically shielded room to minimize environmental noise interference. EEG signals were recorded from 64 scalp electrodes positioned according to the international 10–20 system (Fig. 1). The sampling rate was set to 500 Hz. Reference electrodes were placed on the left and right mastoids, and an additional reference electrode was positioned at the nose tip. Vertical electrooculogram (vEOG) signals[17] were recorded via electrodes placed above and below the left eye to monitor blinks and vertical eye movements. No dedicated horizontal EOG channel was recorded in this dataset. EEG was acquired using Curry Neuroimaging Suite Version 7 (Compumedics Neuroscan, USA), which provided real-time monitoring of signal quality and ensured precise synchronization with stimulus presentation and behavioural responses. To ensure high signal quality, electrode impedances were maintained below 5 kΩ throughout the experiment. Participants were instructed to remain as still as possible during the experiment to minimize movement artifacts and muscle activity. EEG signals, auditory stimuli, and button-press responses were temporally aligned using synchronized trigger signals, ensuring accurate event marking and reliable data analysis. Throughout data acquisition, the quality of the EEG signals was monitored in real time, and electrode connections were adjusted as necessary to maintain stable and reliable recordings. All raw EEG data were stored in a lossless format for subsequent preprocessing and analysis.

**Auditory Stimuli.** The auditory stimuli consisted of spatialized speech mixtures presented at +90° and −90° azimuths, including speech recordings from three male and three female speakers. All raw speech materials were obtained from the publicly available AISHELL-1 Mandarin speech corpus[18], which is openly downloadable through the official repository (https://openslr.org/33/) or via DOI (https://doi.org/10.1109/ICSDA.2017.8384449). This corpus contains high-quality recordings of semantically meaningful sentences spanning daily life, technology, and finance. The stimuli exhibit natural linguistic structures where words are contextually predictable, facilitating the study of attentional modulation in continuous speech perception. And the AISHELL-1 corpus is distributed under an open research license that permits reuse and redistribution for non-commercial academic purposes, no special approval or institutional data-sharing agreements were required. For stimulus construction, speech samples were randomly selected and paired by choosing one male and one female speaker for each binaural mixture. To ensure consistency and experimental control, the selected speech samples underwent preprocessing, including resampling to 16 kHz and root mean square (RMS) loudness normalization[19], standardizing the intensity and duration of audio across speakers. The speech signals were then spatialized using head-related transfer functions (HRTFs)[20] to simulate sound originating from the left (−90°) and right (+90°) locations, creating a binaural auditory scene with competing sound sources. Each spatial channel contained a randomly paired male and female speaker to ensure variability of speech content and the authenticity of attention switching. The stimuli were presented binaurally through headphones at a sound pressure level of 65 dB SPL, with their presentation timing synchronized with the EEG acquisition using E-Prime 3.0 software[21].

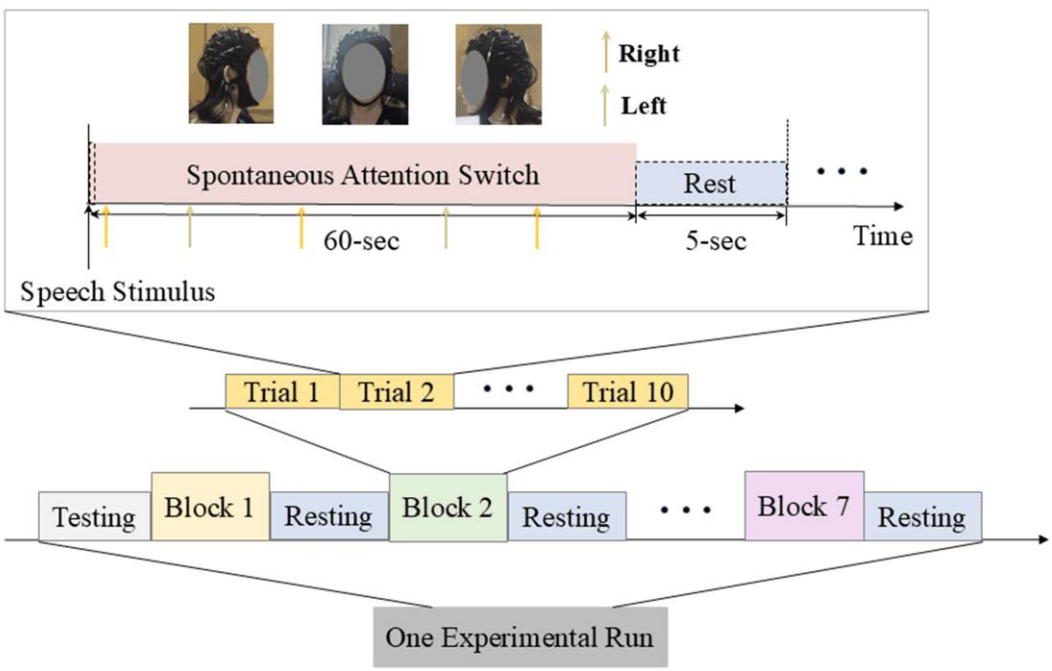


Fig. 2 Experimental procedure and trial structure. A detailed schematic illustrating the trial structure and clearly delineating each phase of the experimental task. The individual depicted in this figure has provided explicit written informed consent for the open publication of her image.

Table 1 Detailed information of the AASD dataset.

| Dataset | # Subjects | Language | # Channels | Block type | Duration per subject | Labels |
|---|---|---|---|---|---|---|

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| | | | | | | |
|---|---|---|---|---|---|---|
| AASD | 18 | Mandarin | 64 | attention-switching blocks | 60 min (6 *10 min/block) | Left (L) or Right (R) |
| | | | | passive listening control block | 10 min (1 *10 min/block) | |

**Experimental protocol.** The experiment consisted of 7 blocks in total, including 6 auditory attention-switching blocks and 1 passive listening control block. In half of the trials, male speech was presented to the left ear and female speech to the right ear. In the other half, the configuration was reversed, presenting male speech to the right ear and female speech to the left ear. This design balanced the spatial distribution of voice gender and sound source location. As illustrated in Fig. 2, the trials were divided into six attention-switching blocks and one passive listening control block, with ten trials per block. The order of blocks was randomized to minimize participants' expectations regarding stimulus sequence. Each trial lasted 60 seconds, during which two spatialized speech streams were simultaneously presented through headphones, creating a competing dual-speaker auditory environment. Participants received no external cues during auditory attention-switching blocks' trial and were instructed to shift their auditory attention freely and spontaneously between the two concurrent speech streams. To record attention switches, participants were provided with two response keys on the keyboard corresponding to "Left" and "Right" attention directions. Importantly, both keys were pressed using the participants' dominant hand rather than using separate hands for each direction, thereby minimizing potential confounds from lateralized motor activations. Button presses and their timestamps were recorded and used to generate attention-switch labels aligned with EEG triggers.

To reduce fatigue and maintain sustained attention, participants were offered up to 10 minutes of rest between every two blocks and a 5-second break between trials. The rest duration was not fixed, and participants could resume the experiment at any time by pressing any key. The entire experimental session lasted approximately 150 minutes, which included EEG setup and calibration, training, the main experiment, and resting periods. Detailed information about the dataset is provided in Table 1. Before the experiment, all participants completed a standardized training session that included an overview of the experimental procedure, practice with the button-press operation, and a demonstration of the auditory attention-switching task. This training ensured that participants fully understood the experimental requirements and labeling procedure, to ensure that they understood the operational norms of attention switching markings.

In addition to the main attention-switching blocks, a passive listening control condition was conducted during the final block. During this block, participants did not perform voluntary auditory attention switching but were instructed to press the response keys alternately at a rate that roughly matched the switching frequency observed during their previous blocks. This condition provides useful behavioural control data for distinguishing motor-related activity from attention-related dynamics.

**EEG Preprocessing.** A standardized preprocessing pipeline was applied to the raw EEG data to ensure signal integrity and minimize artifacts. All preprocessing procedures were performed using EEGLAB v2022[22]. Firstly, conduct a quality check on the original data, eliminating the test trials that have severe noise or equipment abnormalities, to ensure that the subsequent analysis is based on high-quality signals. The EEG data were then re-referenced to the average of the left and right mastoids to reduce global noise and improve spatial resolution. After re-referencing, a finite impulse response (FIR)[23] band-pass filter with a frequency range of 0.1–45 Hz was applied to reduce low-frequency drifts and high-frequency noise. The filtered data were subsequently downsampled to 128 Hz, which reduced computational load while preserving sufficient temporal resolution for attention-switching analyses. Following these preprocessing

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steps, the EEG data were segmented according to individual experimental trials. Event trigger information was then temporally aligned and verified for consistency to ensure precise synchronization between attention-switch labels and the EEG signals. This dataset makes the original data and preprocessed data publicly available. Researchers can select the appropriate data type according to their research needs to support repeated experiments and the flexible application of various analysis methods.

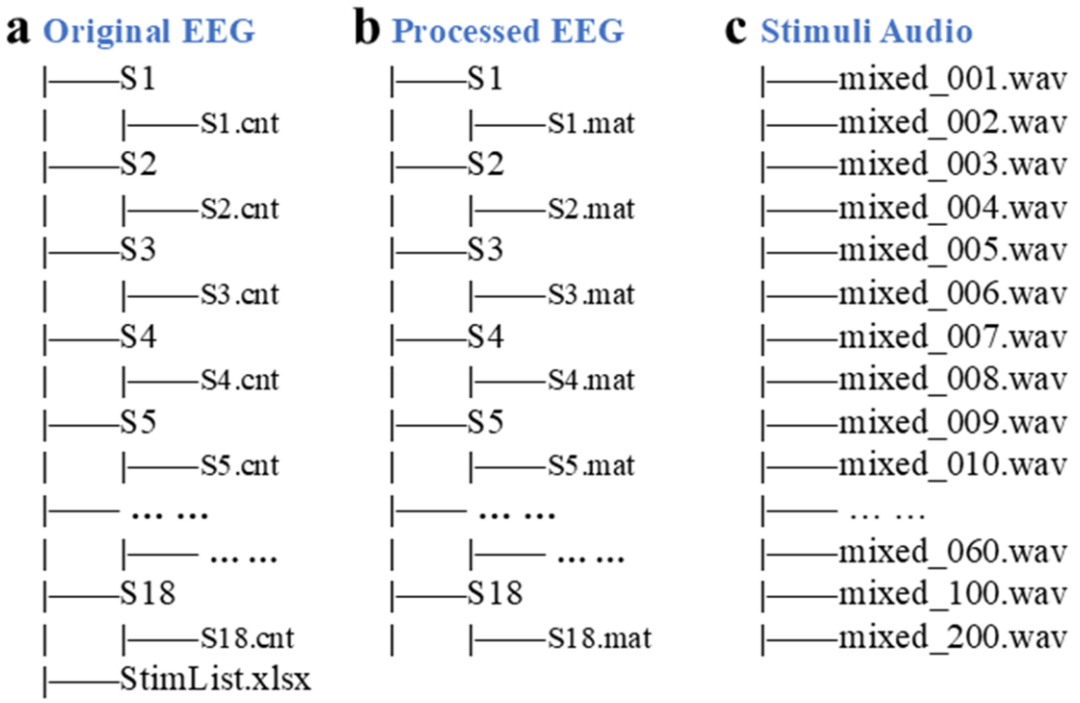


Fig. 3 Structure of the AASD dataset. (a) Original EEG: Raw EEG data in CNT format. (b) Processed EEG: Preprocessed EEG data in MAT format, similarly organized to the original EEG. (c) Stimuli Audio: Spatialized two-channel male and female speech stimuli in WAV format.

## Data Records

The dataset is publicly available on the Zenodo platform[16]. The raw EEG recordings have been organized and stored in a standardized structure, and have been publicly released via https://doi.org/10.5281/zenodo.17413336. As shown in Fig. 3, the dataset is organized into three main folders corresponding to different data types: Original EEG, Processed EEG, and Stimuli Audio, along with an Excel file specifying the correspondence between audio stimuli and trigger events.

- **Original EEG (Fig. 3a)**: Contains unprocessed EEG data in CNT format, without any additional preprocessing.
- **Processed EEG (Fig. 3b)**: Contains preprocessed EEG data in MAT format. The preprocessing steps are identical to those described in the "EEG Preprocessing" section. Each file stores structured EEG signals and event information, where EEG data are saved as a three-dimensional matrix and events are stored as a cell array.
- **Stimuli Audio (Fig. 3c)**: Contains spatialized two-channel male and female speech stimuli in WAV format.

The dataset also includes the EEG and behavioural recordings from the passive listening control condition,

enabling researchers to perform motor-control analyses if desired. This control condition is particularly important for studies that aim to detect attention-switch instances or their timing, because the switch labels in the main task are temporally linked to button presses and may contain motor-related activity in addition to attention-related neural signals. Future users are encouraged to explicitly consider this issue when designing decoding pipelines and interpreting switch-detection results. All EEG files follow the naming convention “S{subject ID}”, and all audio files follow the naming convention “mixed_{Number}”.

## Technical Validation

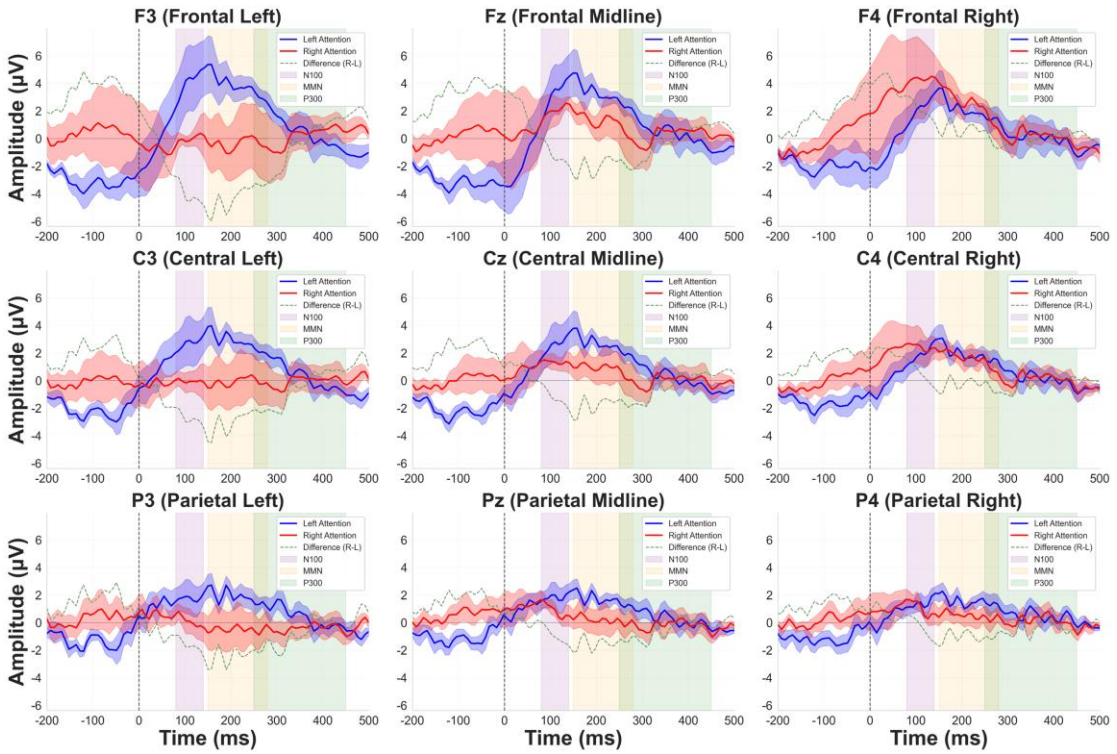


Fig. 4 Grand-average event-related potential (ERP) waveforms recorded during auditory attention switching at nine scalp electrodes (F3, Fz, F4, C3, Cz, C4, P3, Pz, P4). Blue and red traces correspond to left attention and right attention conditions, respectively. Shaded regions indicate the time windows of N100 (80–140 ms), MMN (150–280 ms), and P300 (250–450 ms) components.

This study employed an endogenous auditory attention switch paradigm without explicit external cues. Participants were required to spontaneously switch their auditory attention between left and right sound sources without relying on any exogenous cues. Throughout the experiment, participants repeatedly performed two directions of attentional switching, from leftward attention to rightward attention and from rightward attention to leftward attention. Based on the recorded attentional focus after each completed switch, trials within the corresponding time window were labeled and classified as either leftward attention (Left Attention, LA) or rightward attention (Right Attention, RA). Fig. 4 illustrates the grand-average event-related potential (ERP) waveforms recorded from nine scalp electrodes closely associated with auditory attention (F3, Fz, F4, C3, Cz, C4, P3, Pz, and P4). ERP waveforms for the LA condition are shown in blue, those for the RA condition in red, and the difference waveform (RA − LA) is depicted by a green dashed line. Shaded regions indicate predefined time windows commonly associated with attention-related ERP components, including the N100[24] (80–140 ms), MMN[25] (150–280 ms), and

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P300[26] (250–450 ms). All waveforms were time-locked to the moment of attention switches, which was defined as 0 ms.

For statistical analysis, attention switch direction (LA vs. RA) was treated as a within-subject factor. Two-level repeated-measures analyses of variance (ANOVA) were conducted on mean ERP amplitudes within each time window at each electrode (n = 18, df = 1,17). The significance threshold was set at $\alpha = 0.05$. The selected electrodes covered left, midline, and right sites across frontal, central, and parietal regions[27]. The results revealed a clear asymmetry and regional specificity in the scalp distribution of attention switch direction effects. Compared with right-hemisphere electrodes, left-hemisphere and midline electrodes exhibited more stable and consistent differences between attention conditions across multiple time windows. In the frontal region, the left frontal electrode F3 and the frontal midline electrode Fz showed significantly higher mean ERP amplitudes in the LA condition than in the RA condition (F3: $F(1,17) = 12.23$, $p<0.0005$; Fz: $F(1,17) = 9.14$, $p = 0.007$). In contrast, the right frontal electrode F4 exhibited a difference in the same direction but did not reach statistical significance ($F(1,17) = 2.63$, $p = 0.123$). These findings indicate that frontal cortical activity following spontaneous attention switching is sensitive to attention switch direction, with effects predominantly expressed at left frontal and midline sites[28]. In the central region, the left central electrode C3 and the central midline electrode Cz also showed significantly higher mean amplitudes in the LA condition compared with the RA condition (C3: $F(1,17) = 5.87$, $p = 0.026$; Cz: $F(1,17) = 6.04$, $p = 0.025$). No significant attention switch direction effect was observed at the right central electrode C4 ($F(1,17) = 2.83$, $p = 0.110$). This pattern suggests that attention-related neural differences extend beyond frontal control regions to central areas involved in sensory integration and response preparation[29]. In the parietal region, attention direction showed modest but statistically significant effects at electrodes P3 ($F(1,17) = 5.16$, $p = 0.036$), Pz ($F(1,17) = 5.27$, $p = 0.034$), and P4 ($F(1,17) = 4.51$, $p = 0.048$). Although the effect sizes in the parietal region were generally smaller than those observed in frontal and central regions, the direction of the effects was consistent across participants. These results indicate that parietal regions participate in spatial attentional processing following spontaneous attention switching, although their contribution appears less pronounced than that of frontal and central regions[30].

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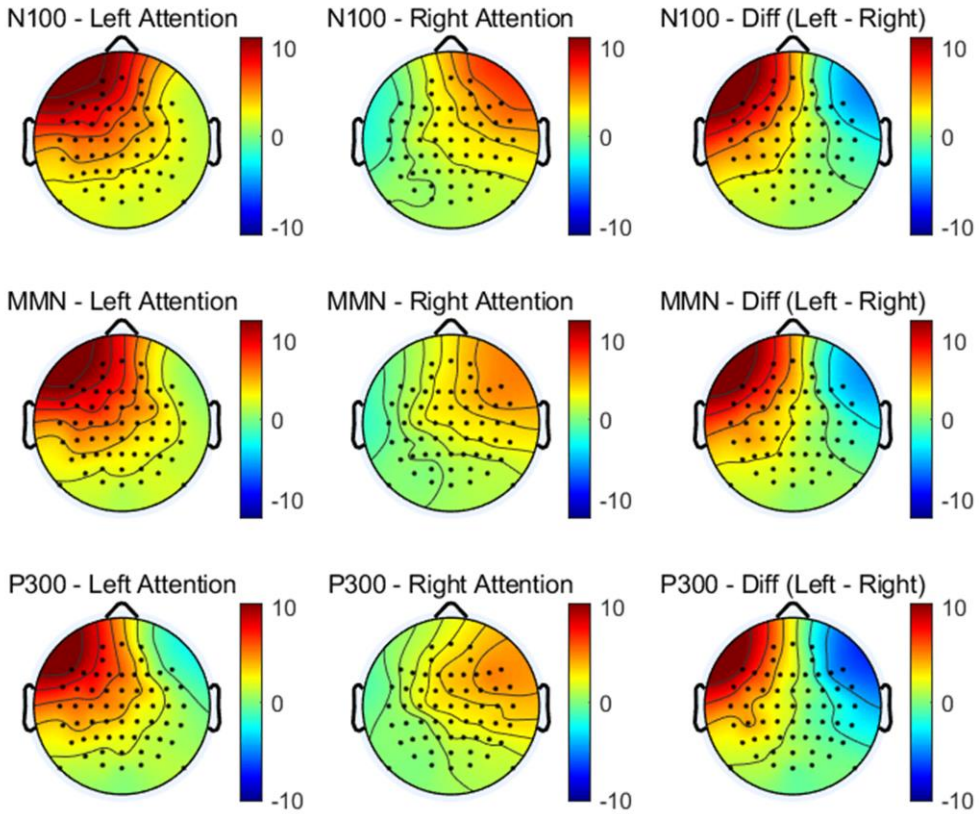
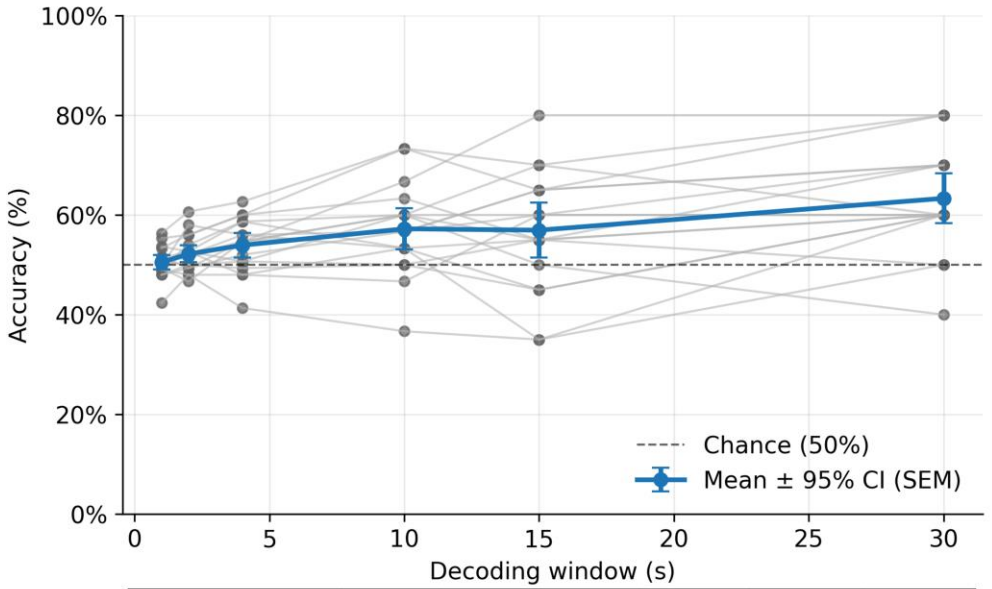



Fig. 5 Topographic maps of ERP components N100 (80–140 ms), MMN (150–280 ms), and P300 (250–450 ms) under left and right attention conditions and their differences.

The results across the nine electrodes demonstrate a clear spatial pattern of attention switch direction effects. Overall, the effects were strongest in the frontal region, followed by the central region, and were relatively weaker in the parietal region. In terms of lateralization, left-hemisphere and midline electrodes showed more robust and consistent effects than right-hemisphere electrodes. The scalp topography results in Fig. 5 consistently demonstrate lateralized distribution patterns related to attention direction, which align with the waveform results reported above. Across the three labelled time windows, the left fronto-central region exhibited more pronounced positive potentials under the LA condition, while the right fronto-central region showed relatively weaker activity. The difference topography (LA−RA) revealed a stable positive difference over the left fronto-central region and a negative difference over the right fronto-central region, indicating a contralateral distribution pattern in the fronto-central regions. Specifically, the left fronto-central area showed enhanced activity during LA, whereas the right fronto-central area showed enhanced activity during RA. This lateralized pattern was observed across all three-time windows, although the spatial extent and intensity varied across temporal stages, suggesting that direction-specific modulations associated with spontaneous attention switching are not confined to a single processing stage but extend across multiple stages following the attention switch[31,32].




| | 1 s | 2 s | 4 s | 10 s | 15 s | 30 s |
|---|---|---|---|---|---|---|
| **Accuracy (%)** | 50.5 | 52.2 | 53.9 | 57.2 | 56.9 | 63.3 |

Fig. 6 TRF attention decoding accuracy under different window length. Gray dots and connecting lines represent individual-subject accuracies across windows. The blue curve shows the group mean at each window, with error bars indicating the 95% confidence interval estimated from the standard error of the mean (Mean ± 95% CI (SEM)). The dashed horizontal line marks chance-level performance (50%).

To evaluate the quality and usability of the AASD dataset, we included a backward temporal response function (TRF) analysis as a stimulus-based decoding baseline validation[33], together with a Transformer-based model as an additional deep-learning benchmark. We first assessed the dataset using a backward TRF decoding framework under dynamic attention-switching conditions. In this approach, EEG signals were used to reconstruct the speech envelope of the attended stream, and the reconstructed envelope was then compared with the envelopes of the left and right speech streams. Attention was classified according to which stream showed the stronger correlation with the reconstructed envelope. Because this dataset involves spontaneous attention switching, the attended stream could alternate between the left and right channels over time. Accordingly, the attended target at each moment was defined based on the switch labels derived from the recorded button presses. For the TRF implementation, a linear ridge-regression model was used to map EEG signals to the attended speech envelope. EEG data were aligned at 128 Hz, and the decoder employed a time-lag window from 0 to 500 ms, corresponding to 65 lags at the given sampling rate. The ridge regularization parameter was set to $\alpha = 100$. The speech representation was defined as the temporal envelope of the stimulus, computed as the absolute value of the Hilbert transform of the audio signal and subsequently filtered using a fourth-order 1–9 Hz band-pass filter.

As shown in Fig. 6, TRF decoding performance improved as the decision window increased, consistent with the well-established behavior of stimulus-based auditory attention decoding methods. The mean classification accuracies across subjects were 50.5%, 52.2%, 53.9%, 57.2%, 56.9%, and 63.3% for decoding windows of 1 s, 2 s, 4 s, 10 s, 15 s, and 30 s, respectively. Performance at longer windows yielded progressively higher decoding accuracy. While this trend is consistent with prior work, the overall decoding accuracies observed in this study, particularly for longer decision windows, are lower than those typically reported in sustained-attention auditory decoding paradigms. This difference is likely attributable to the dynamic nature of the present attention-switching task, in which attentional states

vary over time and neural responses are consequently less stationary. In addition, the use of behaviorally defined switch labels, derived from button presses, may introduce motor-related activity and transitional dynamics that further affect decoding performance. Despite these challenges, the observed increase in performance with window length indicates that the AASD dataset contains meaningful neural information related to selective auditory attention. Overall, the TRF results provide an important baseline validation of the dataset under a more challenging and ecologically valid attention-switching condition.

In addition to the TRF-based baseline, we further evaluated the dataset using a Transformer-based architecture[34,35] as a reference deep-learning benchmark. This analysis was used to examine whether the dataset contains sufficient information to support direct decoding of attentional state in a competing two-speaker environment. The model was designed to detect participants' attentional states (attending to the left or right audio stream). The results demonstrated that the Transformer model effectively captured neural signatures associated with attention switching across decision windows of 0.5 s, 1 s, and 2 s. Notably, at a 1 s decision window, the model achieved an accuracy (ACC) of 70.36% and an F1 score of 62.04%. These findings confirm that the AASD dataset can reliably reflect the dynamic neural activity underlying spontaneous auditory attention switching.

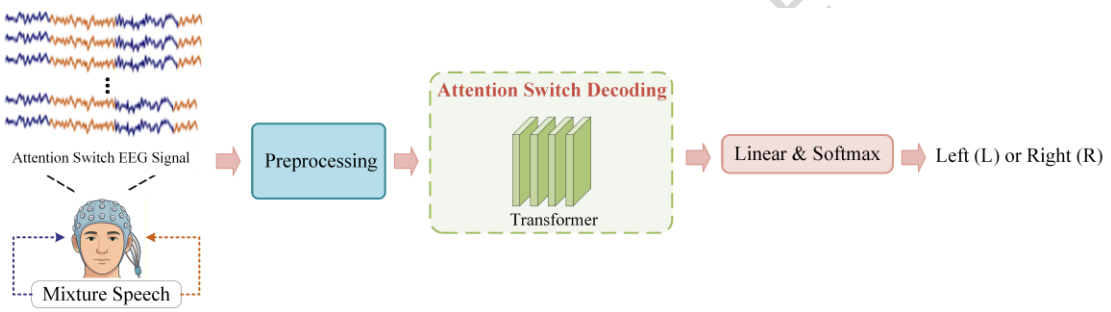



Fig. 7 Model architecture for auditory attention decoding from EEG signals. EEG signals recorded during attention switching tasks are preprocessed and then passed into a Transformer network to learn temporal–spatial representations. A linear and softmax layer generates the final classification of attention direction (left or right).

To decode attention switching states from the preprocessed EEG signals, we used the Transformer architecture illustrated in Fig. 7. The model contained 4 stacked Transformer encoder layers, and each layer included 8 self-attention heads together with a position-wise feed-forward network of hidden dimension 256. Residual connections and layer normalization were applied within each block to stabilize training, and a dropout was used to reduce overfitting. After feature extraction by the Transformer encoder, the resulting representation was aggregated by global average pooling and passed to a fully connected classification layer with softmax activation to predict the attentional state. The key advantage of this model lies in its ability to capture global spatiotemporal dependencies in EEG signals through self-attention mechanisms. The encoder employs multi-layer self-attention and feed-forward networks to extract high-dimensional spatiotemporal features and suppress noise. The resulting representation is then aggregated and passed to a fully connected classification head to generate binary attention-state predictions. Compared with conventional CNNs[36] focused on local feature extraction and RNNs[37] had limited capacity for long-term temporal modeling, the Transformer's global modeling ability makes it particularly well-suited for analyzing attention switching processes that require cross-region and cross-temporal integration.

An 8:1:1 ratio was used to split the data into training, validation, and test sets at the trial level. All EEG windows extracted from the same trial were assigned to the same partition, such that each trial

contributed exclusively to either the training, validation, or test set. To minimize overfitting, a 5-fold cross-validation strategy was employed. Model training was conducted using the PyTorch framework[38] with the Adam optimizer[39] (learning rate = 0.0001, batch size = 16). The model was trained for 50 epochs to ensure convergence. Cross-entropy loss[40] was used as the objective function, and dropout (0.3) along with weight decay (0.0001) was applied to further reduce overfitting.

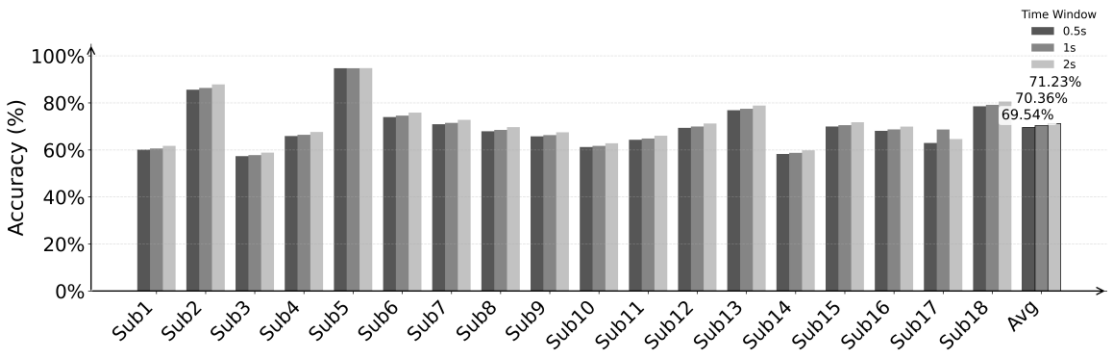



Fig. 8 Accuracy performance of the Transformer model across 18 subjects and the average value at different time windows (0.5 s, 1 s, and 2 s).

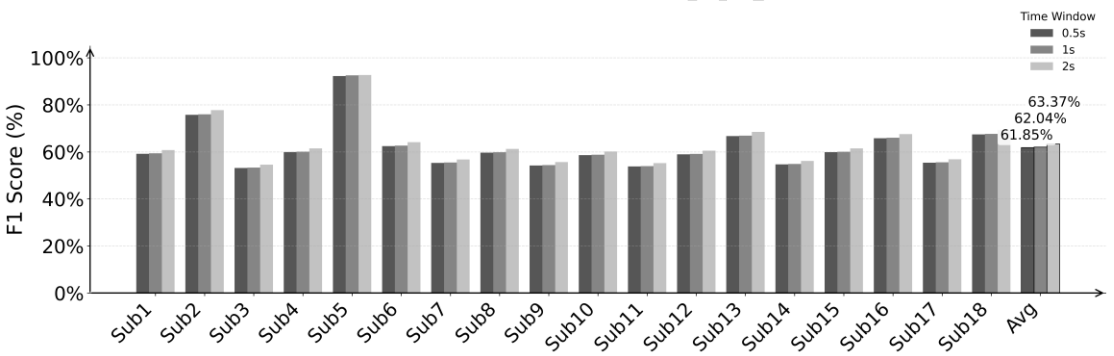



Fig. 9 F1 score performance of the Transformer model across 18 subjects and the average value at different time windows (0.5 s, 1 s, and 2 s).

Two key metrics were used to evaluate decoding performance, complemented by an analysis of brain region feature consistency to validate the physiological plausibility of the data:

- **Accuracy (ACC)**: Measure the alignment between the attention states predicted by the model and the actual attention states of the participants. Higher ACC indicates better decoding performance and serves as a fundamental metric for classification tasks[41,42].
- **F1 score**: Integrates precision and recall to mitigate potential bias from class imbalance. Higher F1 values reflect greater model robustness[43].

As shown in Fig. 8 and Fig. 9, the Transformer achieved robust decoding performance on the spontaneous auditory attention switching dataset. The average accuracy across decision windows of 0.5 s, 1 s, and 2 s were 69.54%, 70.36%, and 71.23%, respectively, all significantly higher than the chance level of 50%. These results demonstrate that the dataset contains sufficient discriminative neural features related to attention switching.

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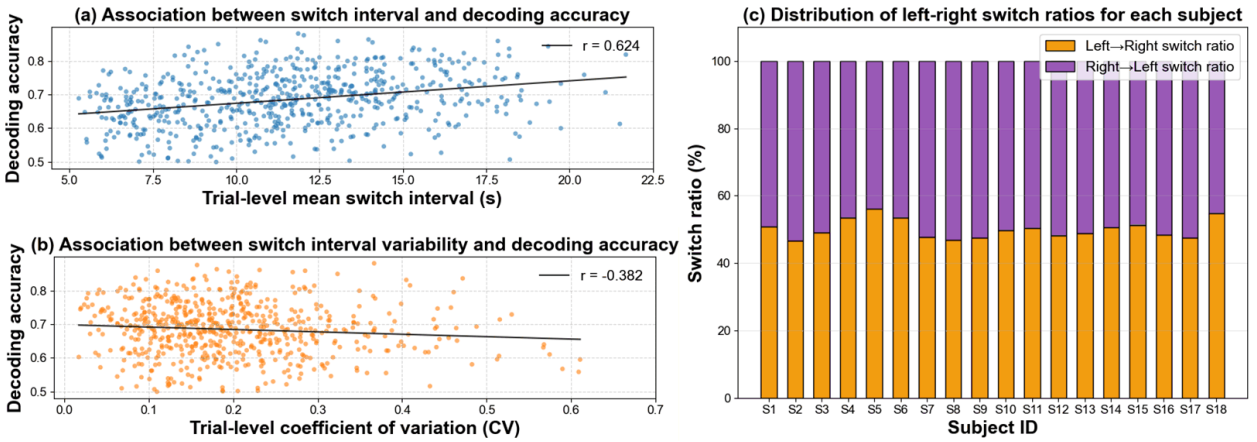


Fig. 10 Scatterplots showing the associations between attention switch interval and decoding accuracy (a), and between switch interval variability (coefficient of variation, CV) and decoding accuracy (b). Each point in panels (a) and (b) represents a trial, and bars in panel (c) represent subject-wise switch ratios.

In addition to the EEG recordings and auditory stimuli, this dataset also includes behavioral data collected during a spontaneous auditory attention-switching task. Based on key press timestamps, we extracted two types of behavioral metrics: (1) Switch Count, representing the number of bidirectional attention switches; and (2) Switch Interval, defined as the temporal duration between successive keypresses. These metrics provide a quantitative basis for interpreting the inter-subject variability observed in decoding performance. Fig. 10 illustrates the relationship between trial-level switch interval characteristics and decoding accuracy. Fig. 10 (a) demonstrates a positive correlation between switch interval and decoding accuracy (r = 0.624). As shown in the figure, data points are primarily distributed within the interval range of 7.5 to 15.0 seconds, with accuracy increasing as the interval lengthens. This suggests that longer switch intervals provide more stable and continuous neural state windows, facilitating the establishment of more reliable decision boundaries by the decoding model. Trials with shorter switch intervals (< 7.5 s) typically correspond to lower accuracy, likely due to frequent switching causing rapid fluctuations in neural states, which limits the model's ability to maintain stable representations. Fig. 10 (b) reveals a negative correlation between the coefficient of variation (CV) of switch intervals and decoding accuracy (r = -0.382). The coefficient of variation is defined as the ratio of the standard deviation to the mean of all switch intervals within a trial (CV = standard deviation/mean), serving as a metric that quantifies the stability of switch intervals. As illustrated, data points are predominantly distributed within the CV range of 0.1 to 0.4, with accuracy decreasing as CV increases. Trials with lower CV values (< 0.2) generally exhibit higher accuracy, indicating that the stability of switching temporal patterns significantly influences decoding performance. These trials demonstrate more uniform distributions of switch intervals, providing more consistent temporal structures that facilitate the model's capture of stable neural patterns. Conversely, trials with higher CV values (> 0.4) typically correspond to lower accuracy, possibly due to irregular switching times introducing temporal inconsistencies that disrupt the continuity of neural states, compromising decoding stability. These findings collectively indicate that both longer switch intervals and lower temporal variability are beneficial for improving decoding accuracy. Longer intervals provide extended analysis windows, while lower variability ensures consistency in temporal structure. Additionally, Fig. 10 (c) highlights the differences in left-right switch ratios across subjects, showing that most subjects have a fairly balanced number of left-to-right and right-to-left attention switches.

In summary, the present technical validation suggests that the AASD EEG dataset provides reliable neural and behavioural data for studying spontaneous auditory attention switching. The TRF decoding and Transformer-based decoding results show that the dataset contains discriminative information relevant

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to attentional state, while the ERP findings are consistent with established neurophysiological patterns associated with auditory attention and attentional reorientation. Together, these results support the overall quality and usability of the dataset for future research on the neural mechanisms of spontaneous auditory attention switching and for the development and evaluation of decoding approaches. At the same time, users should be aware that the annotated switch times are behaviourally defined by key presses and may be temporally confounded by motor-related activity, particularly in studies that aim to detect switch events directly from EEG. To facilitate such analyses, the dataset includes a passive listening control block that can be used to help dissociate motor-related components from attention-related neural processes.

## Code Availability

The code for importing and processing the data described in this paper is publicly available on GitHub (https://github.com/XXuefeii/AASD-Processing).

## Data Availability Statement

The Auditory Attention Switching Dataset (AASD) is publicly available on the Zenodo platform at the following DOI: https://doi.org/10.5281/zenodo.17413336.

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## Author contributions

Xuefei Wang: Conceptualization, Collected the data, Investigation, Methodology and Writing - Original Draft Preparation; Yuting Ding: Validation, Resources; Yueting Ban: Performed the experiment, Collected the data; Lei Wang: Formal analysis; Fei Chen: Supervision, Writing - Review & Editing.

## Competing interests

The authors declare no competing interests.